EuroHaptics Workshop on Haptic Standards and Interoperability

A Study of Deadzone Structure for Multi-Dimensional Haptic Force Perception

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INTRODUCTION

- A perceptual adaptive sampling approach, based on Weber's law of perception, has been used to reduce haptic data in TPTA systems.

$$\frac{|F_n - F_{n-1}|}{|F_{n-1}|} \geq \delta$$

- 1D perceptual deadband: $(1-\delta)F_{n-1} < F_n < (1+\delta)F_{n-1}$

Objective:

Structure of 2D perceptual deadzone

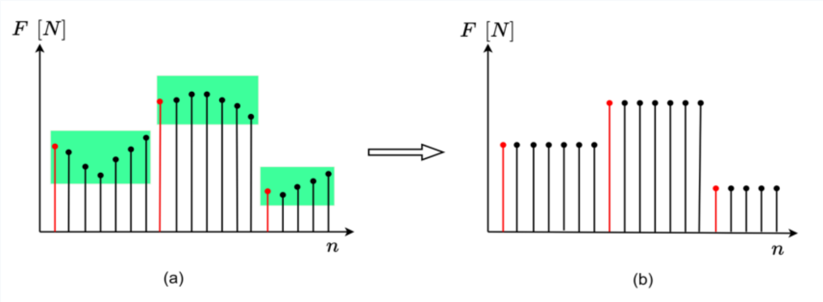

Fig. 1: An illustration of the 1D perceptual deadband approach at Transmitter side: green region denotes the perceptual deadband
Receiver side: Signal reconstruction using the sample-and-hold technique.

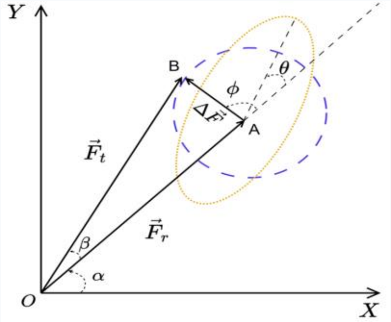

Fig. 2: A possible 2D perceptual deadzone structure around the 2D reference force stimuli

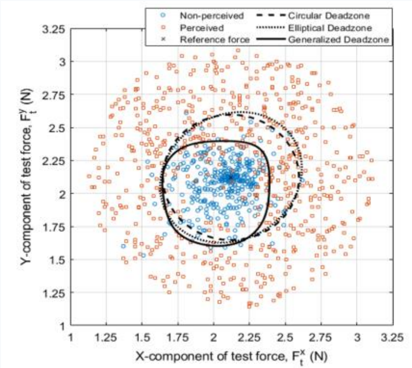

Fig. 3: Scatter Plot for one participant

METHODOLOGY

Machine learning classification-based framework

Setup & Stimuli:

PHANToM Premium 1.5 HF (2D haptic force stimulus +XY-plane).

$$\vec{F}_r = 3\angle 45^\circ, \Delta\vec{F} = [0,1]N, \phi = [0^\circ, 360^\circ]$$

Data Collection: 12 users, 1000 trials per user

Paradigm: Participant is exposed to a 2D kinesthetic force stimulus and is asked to react to the change.

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RESULTS AND ANALYSIS

Shape:

- Area is statistically not different
- Avg. Errors: E_c, E_e, E_g are 28.98%, 21.03%, 16.68%
- Generalized deadzone is the optimal

Effect of Direction: Significant effect of force direction (ϕ) on JND

Symmetric Behavior: Asymmetric with respect to the reference force

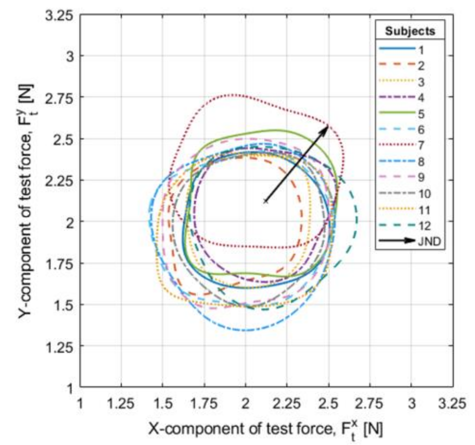

Fig. 4: Optimal boundaries of the generalized deadzone

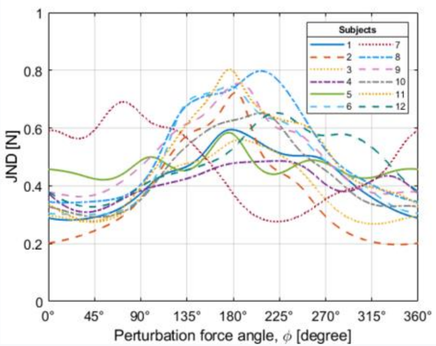

Fig. 5: Force JND vs perturbation force angle (ϕ)

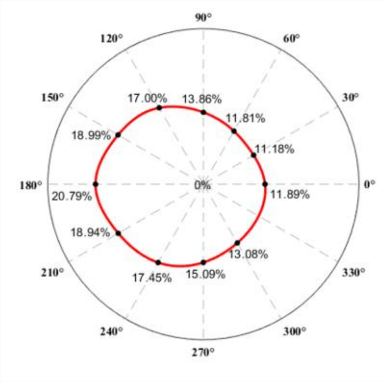

Fig. 6: Average Weber fraction

DISCUSSION & CONCLUSION

- 2D Deadzone is direction variant and radially asymmetric
- δ min: in the reference force direction, δ max: along the perturbation direction (ϕ = 180°)
- The optimal deadzone structure also shows that humans are more sensitive to force perception if the change is made along the reference force direction.
- These findings may support future perceptually adaptive haptic communication standards, coding/compression approaches, and interoperability frameworks by enabling direction-aware force representation and transmission.